

ASHUTOSH ROY

AI Agents & Generative AI Engineer

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SUMMARY

AI Agents & Generative AI Engineer with **1+ year of hands-on experience** building AI agents, LLM-powered applications, and RAG systems. Experienced in developing multi-agent workflows using **LangChain, LangGraph, MCP, FastAPI, and vector databases**. Skilled in **LLM fine-tuning, prompt engineering, AI automation, and deploying production-ready GenAI applications**. Published researcher with experience at **IIT Delhi**, building scalable AI solutions that improve accuracy, reduce manual effort, and automate complex workflows.

EXPERIENCE

Research Intern

Jan 2025 – Jun 2025

Indian Institute of Technology (IIT) Delhi

New Delhi, India

- Built hybrid RAG pipelines using **FAISS, Pinecone, and Apache Solr**, improving retrieval accuracy by **35%** across **10,000+** archaeological documents.
- Developed and deployed **FastAPI** services for real-time LLM inference, reducing response latency by **35%** and enabling scalable AI agent workflows.
- Co-authored the research paper **SARCH: Multimodal Search for Archaeological Archives** with IIT Delhi. arxiv.org/abs/2511.05667

AI/ML Engineer Intern

Oct 2024 – Nov 2024

MitoVoid AI

Remote

- Fine-tuned **Mistral** and **LLaMA** models using **QLoRA** for healthcare AI agents, improving response quality and reducing GPU memory usage.
- Built AI-powered healthcare assistants with tool calling, reducing patient query resolution time by **40%** through better intent routing and automated responses.
- Developed end-to-end GenAI pipelines from data preprocessing and RAG to model deployment using **FastAPI**, enabling production-ready AI applications.

PROJECTS

Skills for Agents Framework | Python - LangGraph - MCP - AI Agents - Open Source

Apr 2026 – May 2026

- Built an open-source framework for reusable AI agent skills, enabling multi-agent workflows with shared memory, tool calling, and modular task execution.
- Designed the **Skills Interoperability Protocol (SIP)**, reducing prompt duplication by **60%** and improving agent scalability through reusable skill modules.

Shopping Assistant Agent | Python - LangGraph - MCP - Claude 3.5 - FastAPI

Feb 2026 – May 2026

- Built a multi-agent shopping assistant using **LangGraph** to automate product search, comparison, recommendations, and review summarization.
- Developed **MCP** servers for secure tool calling with product APIs and databases, reducing response time by **35%** and improving recommendation quality.

Exam-Helper Agentic RAG | Python - LLaMA 3 - LangGraph - FastAPI - FAISS - Pinecone

Dec 2024 – Apr 2025

- Built an Agentic RAG system using **LLaMA 3** and **LangGraph** with support for PDFs, images, and YouTube transcripts.
- Implemented semantic retrieval with **FAISS** and **Pinecone**, reducing hallucinations by **40%** and improving answer accuracy through dynamic context retrieval.

TECHNICAL SKILLS

AI Agents & GenAI: LLMs, AI Agents, Agentic AI, Multi-Agent Systems, RAG, Prompt Engineering, Function Calling, Tool Calling, MCP, LangChain, LangGraph, CrewAI, AutoGen, LlamaIndex

Machine Learning: Fine-Tuning (QLoRA/LoRA), NLP, Deep Learning, Multimodal AI, Speech Processing, Embeddings, Vector Search

Frameworks & Libraries: PyTorch, Hugging Face Transformers, TensorFlow, Scikit-learn, FastAPI, Streamlit

Vector Databases & Search: FAISS, Pinecone, Qdrant, Apache Solr, MySQL

Languages & Tools: Python, SQL, C++, Git, Docker, Linux, VS Code, Google Colab

ACHIEVEMENTS

- Published **2 research papers**, including one accepted at **RECCAP 2026 (IEEE)** and another co-authored with **IIT Delhi**.
- Solved **150+** DSA problems on LeetCode, strengthening data structures and algorithms.
- Won **2 AI/ML Hackathons**, building innovative AI solutions under time constraints.
- Made contributions to **open-source AI/ML projects** on GitHub and Hugging Face.
- Earned certifications in **Machine Learning, Deep Learning, Generative AI, NLP, and Python**.

EDUCATION

Chhattisgarh Swami Vivekananda Technical University (CSVTU)

2022 – 2026

B.Tech (Hons) in Computer Science and Engineering (Artificial Intelligence)

Bhilai, Chhattisgarh

St. Columbus School

2020 – 2021

Higher Secondary (Class XII - Science PCMB) - 87%

Faridabad, Haryana